

Eric Chen

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EDUCATION

University of Southern California

M.S. Electrical Engineering | B.S. Computer Engineering & Computer Science

Aug. 2023 — May 2027

Los Angeles, CA

EXPERIENCE

Development Tools SWE Graduate Intern | Python, Virtuoso, EDA

Jun. 2026 — Present

Intel

Santa Clara, CA

- Developing graph-based algorithms to convert Cadence netlist outputs into structured connectivity features.
- Engineered global-histogram features and random forest models to predict layer assignments from netlist signals.
- Automated design-rule checks to validate model predictions against layout-derived ground truth during iteration.
- Integrated model outputs into analysis scripts to accelerate debugging across chip-design flows.

Technology Intern | Python, SQL, REST APIs, WMS

May 2025 — Aug. 2025

MUJI USA

New York, NY

- Built REST APIs to sync e-commerce orders with the WMS; cut p95 order-sync latency by 42% across NYC stores.
- Optimized inventory/allocations tables and queries, making reports $3.1\times$ faster and cutting p99 latency 61%.
- Built idempotent verification checks; improved online-warehouse consistency and reduced leftover stock issues.
- Wrote daily data integrity scripts (diff scans, duplicates); prevented 1.3k+ bad records/month from reaching reports.

AI Engineer Fellow | LLMs, Docker, CI/CD

Dec. 2025 — Present

Handshake AI

Remote

- Fine-tuning large language models for code generation using supervised learning on open-source codebases.
- Curating gold-standard demonstrations emphasizing functional correctness and multi-file reasoning capabilities.
- Evaluating model outputs against benchmarks for accuracy, code quality, and cross-file dependency handling.
- Built CI/CD pipelines and Dockerized environments for autonomous regression testing across model versions.

Course Producer for CS353 (Intro to Networking)

Spring 2026

University of Southern California

Los Angeles, CA

- Mentored students in CS353 through weekly tutoring hours with structured feedback on algorithms and code quality.
- Collaborated with course staff to clarify material, debug student code, and improve overall learning outcomes.

PROJECTS

Parallel MCCFR for No-Limit Texas Hold'em | CUDA C++, cuRAND, OpenMP, MPI, A100 GPU

Apr. 2026

- Built a poker AI to approximate Nash equilibrium via Monte Carlo CFR self-play and Linear CFR+ (Pluribus).
- Ported to CUDA C++, running 65k simultaneous games per A100 kernel for high-throughput parallel self-play.
- Eliminated Python serialization overhead (57s/worker) via shared-memory merging and GPU-optimized data layouts.
- Deployed a live-play bot using hash-based strategy lookup for sub-microsecond decision latency.

Game Resource Detection System | Python, PyTorch, YOLOv11, OpenCV, NumPy

Jul. 2025

- Annotated a custom multi-class dataset in Roboflow and fine-tuned YOLOv11 for in-game resource detection.
- Built an OpenCV/NumPy augmentation pipeline (HSV jitter, mosaic, blur) to harden rare-class accuracy.
- Trained YOLOv11 in PyTorch with mixed-precision and tuned anchors, hitting real-time inference at native res.
- Packaged as a multi-threaded Python screen-capture overlay with non-blocking inference for live automated detection.

Makeshift Router | C++, POSIX Threads, Sockets, Network Protocol Design

Apr. 2025

- Constructed a multi-threaded router simulation handling concurrent packet forwarding, routing, and TTL expiration.
- Designed a layered protocol stack with checksums, distance-vector routing, and dynamic table updates.
- Implemented socket-based IPC for inter-node communication with custom serialization and timeout handling.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Verilog

Frameworks/Libraries: React, Node.js/Express, Flask, PyTorch, scikit-learn, CUDA, OpenCV, NumPy, WebSockets

Systems/Tools: Linux, Git, Docker, GDB, OpenMP, POSIX Threads, Sockets, Vivado, VS Code, Cadence, Wireshark

Databases/Cloud: PostgreSQL, MySQL, Google Cloud Platform

Other: AutoCAD (Certified) — Mandarin (fluent)

RELEVANT COURSEWORK

Data Structures and Object-Oriented Design; Principles of Software Development; Introduction to Algorithms and Theory of Computing; Introduction to Operating Systems; Computer System Organization; Introduction to Digital Circuits; Professional C++; Parallel and Distributed Computing; MOS VLSI Circuit Design